

STATION NORD, Greenland

WMO: 043120

Lat: 81.600N

Lon: 16.650W

Elev: 118

StdP: 14.63

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-41.4	-38.0	-46.6	0.4	-40.1	-43.4	0.4	-37.5	40.1	4.8	32.8	-0.9	4.5	210	1.124

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	54.4	45.7	51.6	44.1	48.7	42.5	46.2	53.8	44.4	51.0	42.6	48.3	12.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
40.2	36.8	48.6	38.6	34.5	47.3	36.8	32.2	45.6	18.3	54.1	17.4	51.2	16.5	48.5	58.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.4	28.5	22.5	DB	-43.3	58.4	4.3	3.0	-46.4	60.6	-49.0	62.4	-51.4	64.1	-54.5	66.3	(4)
(5)			WB	-44.1	49.5	4.6	3.0	-47.4	51.7	-50.1	53.5	-52.7	55.2	-56.1	57.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-17.3	-18.3	-19.9	-6.9	15.5	33.2	40.1	36.8	20.3	2.6	-8.1	-14.2	(6)	
	DBStd	23.79	11.63	12.91	9.59	9.54	8.85	5.50	6.23	6.05	7.63	9.68	10.05	11.42	(7)	
	HDD50	16261	2086	1912	2166	1708	1069	503	312	410	891	1470	1743	1991	(8)	
	HDD65	21729	2551	2332	2631	2158	1534	953	771	874	1341	1935	2193	2456	(9)	
	CDD50	7	0	0	0	0	0	0	6	1	0	0	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	8.2	7.6	8.0	7.0	7.6	8.1	8.6	8.8	9.3	9.0	7.8	9.0	7.7	(14)	
	PrecAvg	10.3	0.9	0.9	1.0	1.1	0.6	0.3	0.9	0.9	1.4	1.1	1.1	1.1	(15)	
(16) Precipitation	PrecMax	17.2	2.5	3.9	3.9	4.6	2.4	1.5	4.1	2.6	3.7	3.5	3.2	4.1	(16)	
	PrecMin	5.6	0.1	0.0	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	(17)	
	PrecStd	3.0	0.6	0.9	1.0	1.0	0.6	0.4	0.9	0.6	1.1	0.8	0.9	0.9	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	28.5	25.1	26.0	37.8	50.4	59.6	55.5	40.3	32.4	28.6	28.3	(19)	
		MCWB	24.6	27.3	24.3	25.1	33.7	42.7	48.0	45.8	36.4	28.8	27.5	27.0	(20)	
	2%	DB	12.2	20.5	6.2	16.1	34.3	46.3	56.2	52.2	35.6	22.1	18.8	19.0	(21)	
		MCWB	11.5	18.5	5.3	14.8	31.1	40.0	46.4	44.5	32.5	20.4	17.3	17.8	(22)	
	5%	DB	4.1	10.2	-1.9	10.8	31.5	43.8	53.7	49.5	32.3	18.6	10.2	6.8	(23)	
		MCWB	3.3	8.7	-2.6	9.6	29.0	38.6	45.5	42.9	30.6	17.5	9.2	6.0	(24)	
	10%	DB	-1.2	-3.2	-6.4	6.7	28.3	41.4	51.2	47.0	30.5	15.3	5.8	0.3	(25)	
		MCWB	-1.9	-3.8	-7.0	5.6	26.2	37.0	43.9	41.4	28.9	14.4	5.0	-0.4	(26)	
	0.4%	WB	24.7	27.8	24.4	24.8	34.2	43.8	48.7	47.0	36.2	29.6	27.6	27.1	(27)	
		MCDB	25.5	28.8	25.3	26.0	37.4	49.1	57.0	54.0	39.0	32.5	28.6	28.1	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	11.4	18.5	5.3	14.9	31.4	40.7	47.1	44.8	32.8	20.6	17.5	17.8	(29)	
		MCDB	12.1	20.4	6.1	16.2	33.9	45.9	55.1	51.3	34.4	21.8	18.8	19.5	(30)	
	5%	WB	3.4	9.0	-2.6	9.6	28.9	38.6	45.7	43.4	31.0	17.6	9.3	5.9	(31)	
		MCDB	4.0	10.3	-2.0	10.7	31.1	43.2	53.3	48.8	32.2	18.7	10.2	6.7	(32)	
	10%	WB	-1.8	-3.8	-7.0	5.7	26.4	37.1	44.1	41.7	28.9	14.6	5.0	-0.3	(33)	
		MCDB	-1.2	-3.4	-6.5	6.7	28.2	41.3	50.8	46.6	30.3	15.3	5.7	0.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.2	11.0	11.3	9.8	8.3	9.7	8.9	7.7	8.9	10.3	11.2	(35)	
		MCDBR	17.4	19.4	15.7	13.8	12.9	13.4	14.1	13.7	10.8	11.4	13.7	19.5	(36)	
		MCWBR	16.9	18.6	15.5	12.9	10.6	9.4	8.9	9.5	9.4	10.7	13.4	18.8	(37)	
	5% WB	MCDBR	17.2	19.4	15.5	13.7	11.7	13.2	14.4	13.5	10.5	10.9	13.7	19.6	(38)	
		MCWBR	16.7	18.6	15.2	13.0	9.7	9.5	9.3	10.0	9.6	10.2	13.3	18.9	(39)	
(40) Clear-Sky Solar Irradiance	taub	N/A	N/A	0.183	0.261	0.282	0.293	0.290	0.282	0.205	N/A	N/A	N/A	N/A	(40)	
	taud	N/A	N/A	1.989	2.071	2.099	2.143	2.273	2.266	2.065	N/A	N/A	N/A	N/A	(41)	
	Ebn at Noon	0	0	170	232	256	260	256	228	161	0	0	0	0	(42)	
	Edh at Noon	0	0	11	25	32	33	27	21	11	0	0	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg	0	0	180	1051	1903	2331	1960	1158	342	0	0	0	0	(44)	
	RadStd	0	0	18	89	166	191	135	75	32	0	0	0	0	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page